

CATL 电池储能系统 ModBus TCP 通信协议

ModBus TCP (ETH↔EMS)

CATL PMT
State: (In Analysis)

Update Date:
2022年7月27日
Version:V1.3

Catalogue

System overview.....	4
Communication protocol	4
Communication baud rate.....	4
Physical layer.....	4
TCP message format.....	4
Address distribution.....	5
Application layer message format	5
Data format definition.....	6
BMU address distribution rules.....	7
BMU message detailed definition.....	7
MBMU register address definition	7
Message data format example.....	14
Error frame format.....	15
Appendix I	16
Appendix II	17
Appendix III	17
Appendix IV	18

NO.	Version	Date	Author	Audit	Changed Contents
1	1.0	2022/3/3	ShiHG	zhangx	户外水冷平台、风冷平台及大客户项目等融合协议
2	1.1	2022/4/21	ShiHG	zhangx	1、融合户外水冷平台、风冷平台及大客户项目的附录故障列表 2、寄存器地址错误修改: 0x0040修改为0x003a, 0x0041-0x00ff 修改为0x003b-0x00ff
3	1.2	2022/4/24	ShiHG	zhangx	1、Appendix 0x0001 bit12-0x0002 bit7 增加MBMU干接点故障
3	1.3	2022/4/24	ShiHG	zhangx	1、删除 BMUregister address defina 0x0038 (High voltage command on battery) 、0x0039 (DOD) 、0x003a(High voltage command on battery)等信号, 因工控机项目中上位机的逻辑, ETH暂时不做 2、删除 BMU register address defina 0x0421, 因代码无该信号、为预留状态---预留 3、增加 BMU register address defina 0x042f 信号定义 (退出低功耗理由)——预留 4、删除 BMUregister address defina 0x0034, 因与 0x0304信号定义重复 5、地址定义错误更正: Appendix 0x0001 bit11 信号移至 0x0002 bit11
4	1.4	2022/10/27	HuangKC	WangM	1.增加EMS控制绝缘开启关闭指令

System overview

This protocol defines the data exchange format between the PCS monitoring computer and the on-site monitoring computer of the battery's energy storage system.

The statute text used in this protocol is the standard modbus TCP/IP protocol.

PMS will be host computer, battery system will be guest computer, port no 502.

Both the host computer and the guest computer should clarify their IP addresses.

All of the messages are transmitted through internet format (big endian).

Communication status with can bus must be maintained during communication. Otherwise, abnormal data 02 will be returned.

Communication protocol

Modbus TCP/IP

Communication baud rate

100M self-adjusting

Physical layer

Physical layer:IEEE802.3, RJ45 connector with 5 or above 5 type cables

TCP message format

0~15					16~31			
Source port					Destination port			
Sequence number（32bits）								
	Acknowledgement number（32bits）							
Data offset （4bits）	Reserve	U	A	P	R	S	F	Window （16bits）
	(4bits)	R	C	S	S	Y	I	
		G	K	H	T	N	N	
Checksum（16bits）					Urgent pointers（16bits）			
Options(optional)					Padding			
Data								
.....								

In the ethernet, the maximum transport unit (MTU) is 1500 chars (8 bit length) long. In an IP package, it can transport data up to 1480 chars except the IP head (20 chars). In a TCP package, it can transport data up to 1460 chars except the TCP head (20 chars).

Address distribution

Float name	IP address	Sub-net mask	Gateway	DNS server
Guest computer (monitoring pc)	192.1.1.xxx	255.255.255.0	192.168.1.1	502
Host computer (monitoring server)	192.1.1.xxx	255.255.255.0	192.168.1.1	

Application layer message format

After device receives a frame of TCP message from the ethernet correctly, it will extract the modbus TCP/IP data segment from the frame.

Data segment format is shown as below:

Transaction process identifier	Protocol identifier	Data length	Address	Function code	Byte quantity	Data
2byte	2byte	2byte	1byte	1byte	1byte	Nbyte

1. Transaction process identifier (byte0, byte1) ——0
2. Protocol identifier (byte2, byte3) ——0
3. Length segment (byte4 is low byte, byte5 is high byte) ——it means the overall data length (address+function code+data) following.
4. Address (byte6) ——it means the address of the guest computer being addressed; this protocol specified the address of guest computer is 64h.
5. Function code (byte7) ——order the guest computer being addressed guest to finish required action. Function code as noted in the table below.
6. Data (byte8~xx) ——data field, includes data guest terminal needs to finish some specified function, and includes uploaded data during servicing.

Function code introduction:

Function code (hexadecimal)	Definition	Function description
0x03h	Read multiple registers	Host computer reads data of guest computer
0x10h	Write multiple registers	Host computer modifies data of guest computer
0x06h	Write single register	Host computer modifies data of guest computer

Data format definition

Data type	Ratio factor	Range	Offset	Actual span	Data quantity	Comment
Overall voltage	0.1v/bit	0 ~15000	0	0 ~ 1500v	2byte	
MBMU Charge/discharge current	1a/bit	0 ~ 40000	-20000	-20000 ~ 20000a	2byte	Negative current means PCS is charging the battery system, Positive current means PCS is discharging battery system.
SBMU Charge/discharge current	0.1a/bit	0 ~ 40000	-2000	-2000 ~ 2000a	2byte	Negative current means PCS is charging the battery system, Positive current means PCS is discharging battery system.
MBMU Allowed charge/Discharge current	1a/bit	0 ~ 40000	-20000	-20000 ~ 20000a	2byte	Allow the charge current to be negative and discharge current to be
SBMU Allowed charge/Discharge current	0.1a/bit	0 ~ 40000	-2000	-2000 ~ 2000a	2byte	Allow the charge current to be negative and discharge current to be
Capacity	0.1ah/bit	0 ~ 60000	0	0 ~ 6000ah	2byte	
Energy	0.1kwh/bit	0~60000	0	0~6000kwh	2byte	
SOC	0.1%/bit	0 ~ 1000	0	0 ~ 100%	1byte	
SOH	0.1%/bit	0 ~ 1000	0	0 ~ 100%	1byte	
DOD	0.1%/bit	0 ~ 1000	0	0 ~ 100%	2byte	
Single cell temperature	1℃/bit	0 ~ 250	-50	-50 ~ 200℃	1byte	
Single cell voltage	0.001v/bit	0 ~5000	0	0 ~ 5v	2byte	
Single cell temperature difference	1℃/bit	0 ~ 250	0	0 ~ 250℃	1byte	
Single cell voltage difference	0.001v/bit	0 ~5000	0	0 ~ 5v	2byte	
SOC difference	0.1%/bit	0 ~ 1000	0	0 ~ 100%	2byte	
Power	0.1kw/bit	0 ~ 40000	-2000	-2000 ~ 2000kw	2byte	
Power factor	0.01/bit	0 ~ 100	0	0 ~ 1	1byte	
Efficiency	1%/bit	0 ~ 100	0	0 ~ 100%	1byte	
Heartbeats	1/bit	0 ~ 255	0	0 ~ 255	1byte	Both 0~15/0~255 will be ok
Accumulated charge/discharge capacity	0.1kwh/bit	0~4294967295	0	0~429496729kwh	4byte	

Accumulated charge/discharge energy	0.1kwh/bit	0~4294967295	0	0~429496729kwh	4byte	
Environment temperature	1°C/bit	0 ~ 250	-50	-50 ~ 200°C	1byte	
TMS water press	0.1bar/bit	0~250	0	0~25bar	1byte	
Real time-year	1y/bit	0 ~ 254	2000	2000~2254y	1byte	
Real time-month	1m/bit	1~ 12	0	1~12m	1byte	
Real time-day	1d/bit	0 ~ 31	0	0 ~ 31d	1byte	
Real time-hour	1h/bit	0 ~ 59	0	0 ~ 59h	1byte	
Real time-minute	1min/bit	0 ~ 59	0	0 ~ 59min	1byte	
Real time-second	1s/bit	0 ~ 59	0	20 ~ 59s	1byte	

BMU address distribution rules

BMU no	Register address	Byte quantity	Variable description	Comment
MBMU	0x0000-0x3ff	2048	MBMU data field	MBMU occupies 1024 addresses
SBMU01	0x0400~0x07ff	2048	SBMU1 data field	SBMU1 occupies 1024 addresses
SBMU02	0x0800~0x0bff	2048	SBMU2 data field	SBMU2 occupies 1024 addresses
SBMUn	0x(400*n) ~ 0x(400*n+3ff)	2048	SBMUn data field	SBMUn occupies 1024 addresses

BMU message detailed definition**MBMU register address definition**

a) MBMU warning messages(read)

Register address	Data quantity	Description	Comment
0x0000	2	System warning (with degrees)	Detailed in appendix I
0x0001-0x0003	4	System Alarm (without degrees)	Detailed in appendix II
0x0004-0x000f	22	Reserve	Reserve

b)MBMU status messages(read)

Register address	Data quantity	Description	Comment
0x0010	2	System status	000 Initialize 001 Normal 010 Full charge 011 Full discharge 100 Warning status 101 Fault status
0x0011—0x001f	28	Reserve	Reserve

c) MBMU summary messages(read)

Register address	Byte quantity	Description	Comment
0x0020	2	System voltage	Battery system level voltage(outer side)
0x0021	2	System Current	Battery system current Charge/discharge current
0x0022	2	SOC	Battery system current SOC data
0x0023	2	SOH	Battery system current SOH data
0x0024	2	Max cell voltage	Battery system current maximum cell voltage
0x0025	2	Min cell voltage	Battery system current minimum cell voltage
0x0026	2	Avg. cell voltage	Battery system current average cell voltage
0x0027	2	Max cell temperature	Battery system current maximum cell temperature

0x0028	2	Min cell temperature	Battery system current minimum cell temperature
0x0029	2	Avg. cell temperature	Battery system current average cell temperature
0x002a	2	Max charge current allowed	Battery system current allowed maximum charge current
0x002b	2	Max discharge current allowed	Battery system current allowed maximum discharge current
0x002c~0x002d	4	Reserve	Reserve
0x002e	2	Ambient average temperature	Battery system Current ambient average temperature
0x002f	2	Reserve	Reserve
0x0030	2	Ambient humidity	Battery system Ambient humidity
0x0031	2	Reserve	Reserve
0x0032	2	Max allowed charge voltage	Battery system current max allowed charge voltage
0x0033	2	Min discharge voltage allowed (Reserve)	Battery system current min discharge voltage allowed
0x0034	2	Reserve	Reserve
0x0035	2	Reserve	Reserve
0x0036	2	Pos. insulation to GND	Battery system positive insulation resistance to ground
0x0037	2	Neg. insulation to GND	Battery system negative insulation resistance to ground
0x0038-0x00ff	X	Reserve	Reserve
0x0100-0x010f	16	MBMU Software Version	Show in ASCII
0x0110-0x011f	16	MBMU Hardware Version	Show in ASCII
0x0120-0x012f	16	ETH Software Version	Show in ASCII
0x0130-0x013f	16	ETH Hardware Version	Show in ASCII
0x0140-0x02ff	X	Reserve	Reserve

d) ACAN transmit message

Register address	Register address	Description	Variable description	Comment
ACANtransmit	0x0300-0x03ff	512	ACAN data transmit field	For more details-ACAN data transmission

BMS->EMS meaasge(read)

Register address	Byte quantity	Description	Comment
0x0300	2	BMS heartbeat	BMS heartbeats 0-255, updated every 1s
0x0301	2	BMS_poweron	BMS high voltage status 0: Power off ready 1: Power on ready 2: Power on fault 3: Power off fault

0x0302	2	BMS_status	BMS system status 0: Initial status 1: Normal status 2: Full charge status 3: Full discharge status 4: Warning status 5: Fault status
0x0303	2	Reserve	Reserve
0x0304	2	Number of HV connected subsystem	Number of HV connected subsystem
0x0305	2	Reserve	Reserve
0x0306	2	Reserve	Reserve
0x0307	2	System remain charge energy	Charge remaining energy
0x0308	2	System remain discharge energy	Discharge remaining energy
0x0309	2	Reserve	Reserve
0x030a	2	Reserve	Reserve
0x030b	2	Reserve	Reserve
0x030c	2	Reserve	Reserve
0x030d	2	Reserve	Reserve
0x030e	2	E-Stop_status	E-Stop status: 1-fault occur 0-normal
0x030f	2	Max system discharge power allowed	Battery system power allowed maximum discharge power
0x0310	2	Max system charge power allowed	Battery system power allowed maximum charge power
0x0311	2	BMS_limit_charge_hv	System extreme high voltage
0x0312	2	BMS_limit_discharge_hv	System extreme low voltage
0x0313	2	System power	System power
0x0314~0x37f	x	Reserve	Reserve

EMS->BMS数据(write/read)

Register address	Byte quantity	Description	Comment	
0x0380	2	EMS heartbeat	Pcs heartbeats 0-255, updated every 1s	
0x0381	2	EMS_cmd	Command of EMS to control BMS relay 0: Initial 1: Stay status 2: Power on cmd 3: Power off cmd	
0x0382	2	EMS_status	EMS system status (reserve)	
0x0383	2	CRC_c2b_rtc	CRC check	
0x0384	2	EMS_RTC_year	EMS time-year	
0x0385	2	EMS_RTC_month	EMS time-month	
0x0386	2	EMS_RTC_date	EMS time-day	
0x0387	2	EMS_RTC_hour	EMS time-hour	
0x0388	2	EMS_RTC_minute	EMS time-minute	

0x0389	2	EMS_RTC_second	EMS time-second
0x038a	2	EMS_Low_Power_cmd	Command of EMS to control BMS enter low power mode
0x038b	2	EMS_insulation_control_cmd	insulation detection off insulation detection 1: Start 2: Turn
0x038c	2	Fault_Clear_cmd	Fault clear command
0x038d-0x03ff	228	Reserve	Reserve

f) SBMU status messages(read)

Register address	Byte quantity	Description	Comment	
0x0410	2	Precharge relay status	0: Open 1: Close	
0x0411	2	Master positive relay status	0: Open 1: Close	
0x0412	2	Master negative relay status	0: Open 1: Close	
0x0413	2	Fan status	0: Not Run 1: Run	
0x0414	2	Balance status	0: Not Run 1: Run	
0x0415—0x041f	X	Reserve	Reserve	

g) SBMU summary messages(read)

Register address	Byte quantity	Description	Comment	
0x0420	2	Battery Subsystem Voltage	Battery subsystem dc outer side voltage	
0x0421	2	Reserve	Reserve	
0x0422	2	Battery Subsystem Current	Battery subsystem current charge/discharge current	
0x0423	2	SOC	Battery system current SOC	
0x0424	2	SOH	Battery system current SOH	
0x0425	2	Max single cell voltage	Battery system current maximum single	
0x0426	2	Min single cell voltage	Battery system current minimum single cell voltage	
0x0427	2	Avg. cell voltage	Battery system current average cell voltage	
0x0428	2	Max single cell temperature	Battery system current maximum temperature	
0x0429	2	Min single cell temperature	Battery system current minimum temperature	
0x042a	2	Avg. single cell temperature	Battery system current average temperature	

0x042b	2	Max allowed charge current	Battery system current maximum allowed charge current
0x042c	2	Max allowed discharge current	Battery system current maximum allowed discharge current
0x042d	2	Reserve	Reserve
0x042e	2	lower power status	lower power status: 0-low power mode 1-normal
0x042f	2	Reserve	Reserve
0x0430	2	Max single cell voltage position	Low byte means cell number, High byte means tank number
0x0431	2	Min single cell voltage position	Low byte means cell number, High byte means tank number
0x0432	2	Max single cell temperature position	Module number
0x0433	2	Min single cell temperature position	Module number
0x0434-0x0436	X	Reserve	Reserve
0x0437	2	Sum of cell voltage	Sum of cell voltage
0x0438-0x043f	16	Reserve	Reserve
0x0440-0x0447	X	Reserve	Reserve
0x0448	2	TMS mode command by BMS	TMS mode command by BMS 0: power off 1: cooling mode 2: heating mode 3: self-circulation mode
0x0449	2	TMS temperature value set by BMS	TMS temperature value set by BMS
0x044a	2	TMS real mode	TMS real mode 0: power off 1: cooling mode 2: heating mode 3: self-circulation mode
0x044b	2	Rack inlet(TMS Outlet) temperature	Rack inlet(TMS outlet) temperature
0x044c	2	Rack outlet(TMS inlet) temperature	Rack outlet(TMS inlet) temperature
0x044d	2	TMS Environment temperature	TMS Environment temperature
0x044e	2	TMS demand power	TMS demand power
0x044f	2	TMS fault code	TMS fault code
0x0450	2	Door state	Door state 0: Close 1: Open
0x0451	2	Fan state in control box	Fan state in control box 0: Stop 1: Run
0x0452-0x047f	X	Reserve	Reserve
0x0452	2	TMS_OutWaterPress	TMS_OutWaterPress
0x0453	2	TMS_BackWaterPress	TMS_BackWaterPress

0x0454-0x0455	4	Historical accumulated charge energy	Historical cumulative charge energy
0x0456-0x0457	4	Historical cumulative discharge energy	Historical cumulative discharge energy
0x0456-0x047f	X	Reserve	Reserve

h) SBMS detail messages(read)

Register address	Byte quantity	Description	Comment	
0x0480~0x06bf	576* 2	Detail message-single cell voltage	All cells: M*N (max 576) M: number of CSC N: number of cells per CSC 1#CSC cells voltage: 0x480 ~ (0x480+N-1) 2#CSC cells voltage: (0x480+N) ~ (0x480+2N-1) M#CSC cells voltage: (0x480+(M-1) * N) ~ (0x480+M*N-1)	
0x06c0~0x077f	192* 2	Detail message-single cell temperature	All cells temperature sensors: M*P (max 192) M: number of CSC P: number of temperature sensor per CSC 1#CSC temperatures: 0x6c0 ~ (0x6c0+P-1) 2#CSC temperatures: (0x6c0+P) ~ (0x6c0+2P-1) M#CSC temperatures: (0x6c0+(M-1) * P) ~ (0x6c0+M*P)	
0x0780~0x07af	48* 2	Detail message-balance information	All cells: M*N (max 576) M: number of CSC N: number of cells per CSC 1#CSC cells balance status: 0x780 ~ (0x780+ceil(N/16) -1) 2#CSC cells balance status: (0x780+ceil(N/16)) ~ (0x780+2* ceil(N/16) -1) M# CSC cells balance status: (0x780+(M-1)* ceil(N/16))~ (0x780+M* ceil(N/16) -1)	
0x07b0~0x07bf	16	Reserve	Reserve	
0x07c0~0x07cf	16	SBMU Software Version	Show in ASCII	
0x07d0~0x07df	16	SBMU Hardware Version	Show in ASCII	

0x07e0~0x07ef	16	CSC Software Version	Show in ASCII ,all CSC belong to this SBMU will be the same Version
0x07f0~0x07ff	16	CSC Hardware Version	Show in ASCII , all CSC belong to this SBMU will be the same Version

Message data format example

If the host computer demands those data saved in 0x0420~0x0432 of the guest computer (19 registers, total 38 bytes), then the messages between host computer and guest computer are shown below:

1. Host computer demands running message of guest computer

Transaction process identifier	Protocol identifier	Data length	System address	Function code	Starting register's address high byte	Starting register's address low byte	Register amount high byte	Register amount low byte
00H 00H	00H 00H	00H 06H	64H	03H	04H	20H	00H	13H

2. Guest computer returns messages to host computer

Transaction process identifier	Protocol identifier	Data length	System address	Function code	Byte number of data	Register 0x0420 high byte of data	Register 0x0420 low byte of data
00H 00H	00H 00H	00H 29H	64H	03H	26H	xxH	xxH

Register 0x0421 high byte of data	Register 0x0421 low byte of data	Register ... High byte of data	Register ... Low byte of data	Register 0x0432 high byte of data	Register 0x0432 low byte of data
xxH	xxH	xxH	xxH	xxH	xxH

Ps: data length means the number of total bytes following the data length bytes, and byte number of data means the bytes' number of the returned registers.

Ps: data length means the number of total bytes following the data length bytes, and byte number of data means the bytes' number of the returned registers.

When host computer demands the same data package from guest computer, it can set communication period according to the importance of the message, and the smallest period should be bigger than 20ms.

Error frame format

Transaction process identifier	Protocol identifier	Data length	System address	Function code	Abnormal code
--------------------------------	---------------------	-------------	----------------	---------------	---------------

00H 00H	00H 00H	Previous data length xx	0	Previous function code H+128	XX in the table below
---------	---------	-------------------------	---	------------------------------	-----------------------

Error code

Abnormal code	Description
0x01	Unsupported function code
0x02	Without can communication
0x03	More than 127 addresses demanded
0x04	Something abnormal happens
0x05~0xff	Reserve

Appendix I

MBMU warning table with degrees

Status bit	Description	Bit	Comment
0x0000 Bit0-bit1	Single cell overvoltage warning	00	Normal
		01	Single cell overvoltage warning1
		10	Single cell overvoltage warning2
		11	Single cell overvoltage warning3
0x0000 Bit2-bit3	Single cell under voltage warning	00	Normal
		01	Single cell under voltage warning1
		10	Single cell under voltage warning2
		11	Single cell under voltage warning3
0x0000 Bit4-bit5	Single cell over temperature warning	00	Normal
		01	Single cell over temperature warning1
		10	Single cell over temperature warning2
		11	Single cell over temperature warning3
0x0000 Bit6-bit7	Single cell under temperature warning	00	Normal
		01	Single cell under temperature warning1
		10	Single cell under temperature warning2
		11	Single cell under temperature warning3
0x0000 Bit8-bit9	SOC low warning	00	Normal
		01	SOC low warning1
		10	SOC low warning2
		11	SOC low warning3
0x0000 Bit10-bit11	Isolation resistance warning	00	Normal
		01	Isolation resistance warning1
		11	Isolation resistance warning3
0x0000 Bit12-bit15	Reserve	-	Reserve

Appendix II

MBMU warning without degrees

Status bit	Description	Bit	Comment
0x0001 bit0	Current over limit warning	0	Normal
		1	Warning
0x0001 bit1	Warning for big voltage difference of single cell	0	Normal
		1	Warning
0x0001 bit2	Warning for big temperature difference within bank	0	Normal
		1	Warning
0x0001 bit3	System overvoltage warning	0	Normal
		1	Warning
0x0001 bit4	System under voltage warning	0	Normal
		1	Warning
0x0001 bit5	Inner communication fault warning	0	Normal
		1	Warning
0x0001 bit6	Cell extreme temperature warning	0	Normal
		1	Warning
0x0001 bit7	Cell extreme voltage warning	0	Normal
		1	Warning

0x0001 bit8	EMS heartbeat fault	0	Normal
		1	Fault
0x0001 bit9	Reserve	-	Reserve
0x0001 bit10	Reserve	-	Reserve
0x0001 bit11	Reserve	-	Reserve
0x0001 bit12	Mains power supply lose warning	0	Normal
		1	Warning
0x0001 bit13	Mains power supply lose fault	0	Normal
		1	Fault
0x0001 bit14	UPS fault	0	Normal
		1	Fault
0x0001 bit15	Fire fighting fault	0	Normal
		1	Fault
0x0002 bit0	Fire fighting warning	0	Normal
		1	Warning
0x0002 bit1	E-Stop press down fault	0	Normal
		1	Fault
0x0002 bit2	E-Stop fault	0	Normal
		1	Fault
0x0002 bit3	Entrance door fault	0	Normal
		1	Fault
0x0002 bit4	Escape door fault	0	Normal
		1	Fault
0x0002 bit5	Main load switch fault	0	Normal
		1	Fault
0x0002 bit6	Lightning protector fault	0	Normal
		1	Fault
0x0002 bit7	Exhaust fan fault	0	Normal
		1	Fault
0x0002 Bit8 - Bit10	Reserve	-	Reserve
0x0002 bit11	Racks numbers of high voltage power on fault	0	Normal
		1	Warning
0x0002 Bit12 - Bit15	Reserve	-	Reserve

Appendix III

SBMU warning with degrees

Status bit	Description	Bit	Comment
0x0400 Bit0-bit1	Single cell overvoltage warning	00	Normal
		01	Single cell overvoltage warning1
		10	Single cell overvoltage warning2
		11	Single cell overvoltage warning3
0x0400 Bit2-bit3	Single cell under voltage warning	00	Normal
		01	Single cell under voltage warning1
		10	Single cell under voltage warning2
		11	Single cell under voltage warning3
0x0400 Bit4-bit5	Single cell overtemperature warning	00	Normal
		01	Single cell overtemperature warning1
		10	Single cell overtemperature warning2
		11	Single cell overtemperature warning3
		00	Normal

0x0400 Bit6-bit7	Single cell undertemperature warning	01	Single cell undertemperature warning1
		10	Single cell undertemperature warning2
		11	Single cell undertemperature warning3
0x0400 Bit8-bit9	SOC low warning	00	Normal
		01	SOC low warning1
		10	SOC low warning2
		11	SOC low warning3
0x0400 Bit10-bit11	TMS fault level	00	Normal
		01	TMS fault level 1
		10	TMS fault level 2
		11	TMS fault level 3
0x0400 Bit12-bit15	Reserve	-	Reserve

Appendix IV

SBMU warning without degrees

Status bit	Description	Bit	Comment
0x0401 bit0	Charge over-current warning	0	Normal
		1	Warning
0x0401 bit1	Discharge over-current warning	0	Normal
		1	Warning
0x0401 bit2	Cell voltage difference warning	0	Normal
		1	Warning
0x0401 bit3	Temperature difference warning	0	Normal
		1	Warning
0x0401 bit4	Cell extreme temperature fault	0	Normal
		1	Fault
0x0401 bit5	Cell extreme voltage fault	0	Normal
		1	Fault
0x0401 bit6	Rack total voltage warning	0	Normal
		1	Warning
0x0401 bit7	Main positive relay close fail fault	0	Normal
		1	Fault
0x0401 bit8	Main negative relay close fail fault	0	Normal
		1	Fault
0x0401 bit9	Main positive relay stuck fault	0	Normal
		1	Fault
0x0401 bit10	Main negative stuck fault	0	Normal
		1	Fault
0x0401 bit11	Auxiliary power fault	0	Normal
		1	Fault
0x0401 bit12	Balance warning	0	Normal
		1	Warning
0x0401 bit13	MCU self-check fault	0	Normal
		1	Fault
0x0401 bit14	Reserve	/	Reserve
0x0401 bit15	Inner communication fault (CCAN)	0	Normal
		1	Fault
0x0402 bit0	Inner communication fault (SCAN)	0	Normal
		1	Fault
0x0402 bit1	Reserve	/	Reserve

0x0402 bit2	Current sensor abnormal fault	0	Normal
		1	Fault
0x0402 bit3	Single cell voltage sampling abnormal	0	Normal
		1	Fault
0x0402 bit4	Module temperature sampling abnormal	0	Normal
		1	Fault
0x0402 bit5	High voltage circuit open fault	0	Normal
		1	Fault
0x0402 bit6	MSD fault	0	Normal
		1	Fault
0x0402 bit7	Rack isolate switch fault	0	Normal
		1	Fault
0x0402 bit8	Rack fuse fault	0	Normal
		1	Fault
0x0402 bit9	Pre-charge Timeout fault	0	Normal
		1	Fault
0x0402 bit10	Extreme over current fault	0	Normal
		1	Fault
0x0402 bit11	Contactor coil abnormal warning	0	Normal
		1	Warning
0x0402 bit12	The warning of different voltage between sampling voltage and sum voltage warning	0	Normal
		1	Warning
0x0402 bit13	MCAN communication warning	0	Normal
		1	Warning
0x0402 bit14	CSC Power Fault	0	Normal
		1	Fault
0x0402 bit15	Current sensor Power Fault	0	Normal
		1	Fault
0x0403 bit0	TMS communication fault	0	Normal
		1	Fault

0x0403 bit1	TMS mode conflict fault	0	Normal
		1	Fault
0x0403 bit2	Fire fault level 1	0	Normal
		1	Fault
0x0403 bit3	Fire fault level 2	0	Normal
		1	Fault
0x0403 bit4	TempSensorState	0	Normal
		1	Fault
0x0403 bit5	SmokeSensorState	0	Normal
		1	Fault
0x0403 bit6	Aerosol state	0	Normal
		1	Happened
0x0403 bit7	Aerosol close fault	0	Normal
		1	Abnormal opened
0x0403 bit8	Aerosol open fault	0	Normal
		1	Abnormal closed
0x0403 bit9	Rack switch state	0	Normal
		1	Happened
0x0403 bit10	Over temperature fault in	0	Normal

0x0403 bit10	control box	1	Over temperature fault in control box
0x0403 bit11	TMS Make BMS Stop	0	Normal
		1	Fault
0x0403 bit12	Fan fault 1#	0	Normal
		1	Fan fault
0x0403 bit13	Fan fault 2#	0	Normal
		1	Fan fault
0x0403 bit14-bit15	Reserve	/	Reserve